InHand Networks

Lightweight Vehicle Tracking, Reliable Connectivity, Precise Positioning

VT200 Vehicle Telematics Gateway

- LTE Cat1 / Cat-M1
- GNSS
- CAN / I/O
- Inertial

1. Product Overview

The InHand VT200 is a vehicle telematics gateway capable of reliable operation even when the vehicle is powered off.

Positioning: LTE + GNSS + inertial navigation telematics gateway for vehicle tracking, fleet management, and asset monitoring

Key Features:

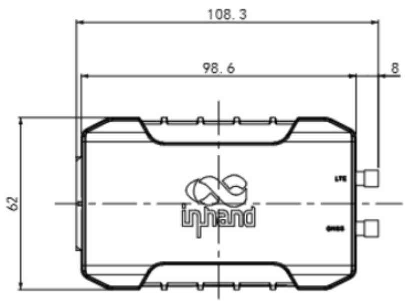
- **High-performance tracking:** GNSS positioning with inertial navigation for continuous location accuracy
- **Driving behavior monitoring:** accelerometer and gyroscope enable real-time monitoring of braking, acceleration, and collision events
- **Extensive interfaces:** CAN bus, RS232/RS485, DI/AI, DO, and 1-Wire for flexible vehicle integration
- **Continuous operation:** cache supports continuous data recording when the internet connection is unavailable
- **Cloud & device ecosystem:** supports mainstream IoT platforms such as AWS/Azure and other tracking services

Core Technical Specifications

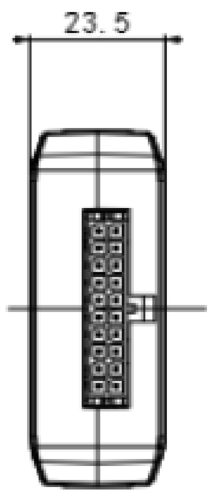
Technical Indicator	Specification
Cellular Network	LTE Cat1 / Cat-M1
Positioning Capability	GNSS (GPS / Galileo / Beidou / GLONASS) + Inertial Navigation (DR)
Cloud Platform Access	Supports AWS IoT, Azure IoT, Aliyun IoT, and mainstream third-party tracking platforms
Transmission Protocols	TCP, UDP, HTTP, MQTT, JT808

Technical Indicator	Specification
Transparent Transmission	Supports transparent transmission for OBD-II / J1939 / CAN / ELD, and RS232/RS485 serial passthrough
Security Capability	Supports SSL/TLS encryption and platform-side security authentication
Dimensions	98.6 × 62 × 23.5 mm (built-in antenna) / 108.3 × 62 × 23.5 mm (external antenna)
Interfaces	CAN, RS232, RS485, DI/AI, DO, 1-Wire, USB Type-C
Power Supply Range	9 ~ 36 VDC
Operating Temperature	-20 ℃ ~ 60 ℃
Protection Rating	IP40
Storage Temperature	-40 ℃ ~ 85 ℃

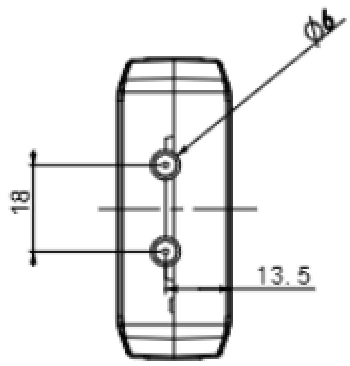
2. Product Dimensions



Front View



Interface Side View



Height Side View

Notes:
1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and **for reference only**.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category / Parameter	Specification
Networking & Cellular	
Network Type	LTE Cat1 / Cat-M1
Cellular Antenna	Built-in ceramic antenna or external SMA (model dependent)
GNSS Antenna	Built-in ceramic antenna or external SMA (model dependent)
SIM	Nano-SIM (4FF), single SIM card
Satellite Navigation	
Satellite Support	GPS / Galileo / Beidou / GLONASS
Channels	72 channels
Initial Positioning Sensitivity	-164 dBm (initial positioning time 26 s)
Tracking Sensitivity	-156 dBm (hot start) / -147 dBm (cold start)
Location Accuracy	2.5 m (CEP50)
Update Frequency	10 MHz
Inertial Navigation	Supports dead-reckoning (DR)

Category / Parameter	Specification
Acceleration	Measurement range: ±2 / ±4 / ±8 / ±16g
Angular Velocity	Measurement range: ±125 / ±250 / ±500 / ±1000 / ±2000dps
Electrical & Environment	
Working Voltage	9 ~ 36 VDC
Operating Temperature	-20 ℃ ~ 60 ℃
Storage Temperature	-40 ℃ ~ 85 ℃
Humidity	95% RH @ 50 ℃ (non-condensing)
ESD	IEC 61000-4-2 (4 kV test)
Protection Rating	IP40
Battery Capacity (battery models)	1000 mAh
Battery Material (battery models)	Lithium-ion
Battery Operating Temperature	Charging: 0 ℃ ~ 45 ℃; Discharging: -20 ℃ ~ 60 ℃
Battery Storage Temperature	-20 ℃ ~ 35 ℃
Mechanical	
Shell Material	ABS + PC
Dimensions	98.6 × 62 × 23.5 mm (built-in) / 108.3 × 62 × 23.5 mm (external)
Vehicle Interfaces	

Category / Parameter	Specification
CAN Bus	1 channel
Ignition Signal	1 channel (ACC/IGT)
Digital Input / Analog Input	4 channels (DI/AI)
Digital Output	2 channels (max. 300 mA)
1-Wire	1 channel
Serial Port	RS232 and RS485
External Interfaces	USB 2.0 Type-C, SIM slot, LED indicators

4. Software Specifications

Category / Parameter	Specification
Network Features	
Cloud Platforms	AWS IoT, Azure IoT, Aliyun IoT, Wialon, Traccar, GPSWox, WhiteLable Tracking, ThingsBoard, MQTT Cloud, Customer Private Cloud, and other supported platforms
Transport Protocols	TCP, UDP, HTTP, MQTT, JT808
Encryption	SSL/TLS
Vehicle Data & Transparent Transmission	
Data Transmission	OBD-II / J1939 / CAN bus / ELD transparent transmission
Serial Transparent Transmission	RS232/RS485 transparent; supports Modbus RTU to Modbus

Category / Parameter	Specification
Event & Reporting	
Event Alarm	Collision detection, motion detection, overspeed, IO change, ignition signal detection
Reporting	Support SMS or FlexAPI over TCP/UDP/MQTT
Configuration Interface	
Configuration	USB 2.0 Type-C
Security	
Authentication	Supported for secure communications (per platform)
Certification	
Certifications*	CE, FCC, IC, PTCRB, E-Mark

5. Ordering Information

Model Rule

Model code: VT200-<WMNN>

<WMNN>: Cellular Type & Module

Product Models

Model	Cellular Type / Region	Bands
VT200-FQ33-BAT	LTE Cat1, North America	LTE-FDD B2/B4/B5/B12/B13; WCDMA B2/B4/B5

VT200-FQ33	LTE Cat1, North America	LTE-FDD B2/B4/B5/B12/B13; WCDMA B2/B4/B5
VT200-FQ33-ANT	LTE Cat1, North America	LTE-FDD B2/B4/B5/B12/B13; WCDMA B2/B4/B5
VT200-FQ33-ANT-BAT	LTE Cat1, North America	LTE-FDD B2/B4/B5/B12/B13; WCDMA B2/B4/B5
VT200-FQ02-BAT	LTE Cat-M1, Global	LTE-FDD B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28/B66/B85
VT200-FQ02	LTE Cat-M1, Global	LTE-FDD B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28/B66/B85
VT200-FQ02-ANT	LTE Cat-M1, Global	LTE-FDD B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28/B66/B85
VT200-FQ02-ANT-BAT	LTE Cat-M1, Global	LTE-FDD B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28/B66/B85

Cable Accessories

Cable	Order Code	Specifications
20 PIN Cable	SCAB000381	P1 is 20PIN female, connected to VT200; P2 is open end, AWG24 for other signals; connector HX30002-20
OBD-II 16 PIN Test Cable	SCAB000399	OBD-II 16 PIN test line; cable standard UL2464; wire length 1500 mm
J1939 6 PIN Test Cable	SCAB000409	J1939 6 PIN interface test line; cable standard UL2464; wire length 1500 mm

6. Contact Us

- **Website:** InHand Networks
- **Copyright:**

7. Connector Pin Assignment

IO 20PIN Definition

Pin	Definition	Description
1	CAN0_H	CAN0 bus high-level signal
2	RS485_A	RS485 A line
3	GND	Signal ground
4	RS232_TX	RS232 transmit signal
5	DO1	Digital output 1
6	GND	Signal ground
7	AI1/DI1	Analog input 1 or digital input 1
8	AI3/DI3	Analog input 3 or digital input 3
9	IGT	Ignition detection input
10	VIN-	Power negative terminal
11	CAN0_L	CAN0 bus low-level signal
12	RS485_B	RS485 B line
13	1Wire	1-Wire bus interface
14	RS232_RX	RS232 receive signal
15	DO2	Digital output 2

16	GND	Signal ground
17	AI2/DI2	Analog input 2 or digital input 2
18	AI4/DI4	Analog input 4 or digital input 4
19	GND	Signal ground
20	VIN+	Power positive terminal